METRIC

MIL-DTL-38999/23C
10 July 2008
SUPERSEDING
MIL-DTL-38999/23B
11 June 2001

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, RECEPTACLE, JAM-NUT MOUNTING, HERMETIC, HERMETIC SOLDER CONTACTS, SERIES III, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and MIL-DTL-38999.

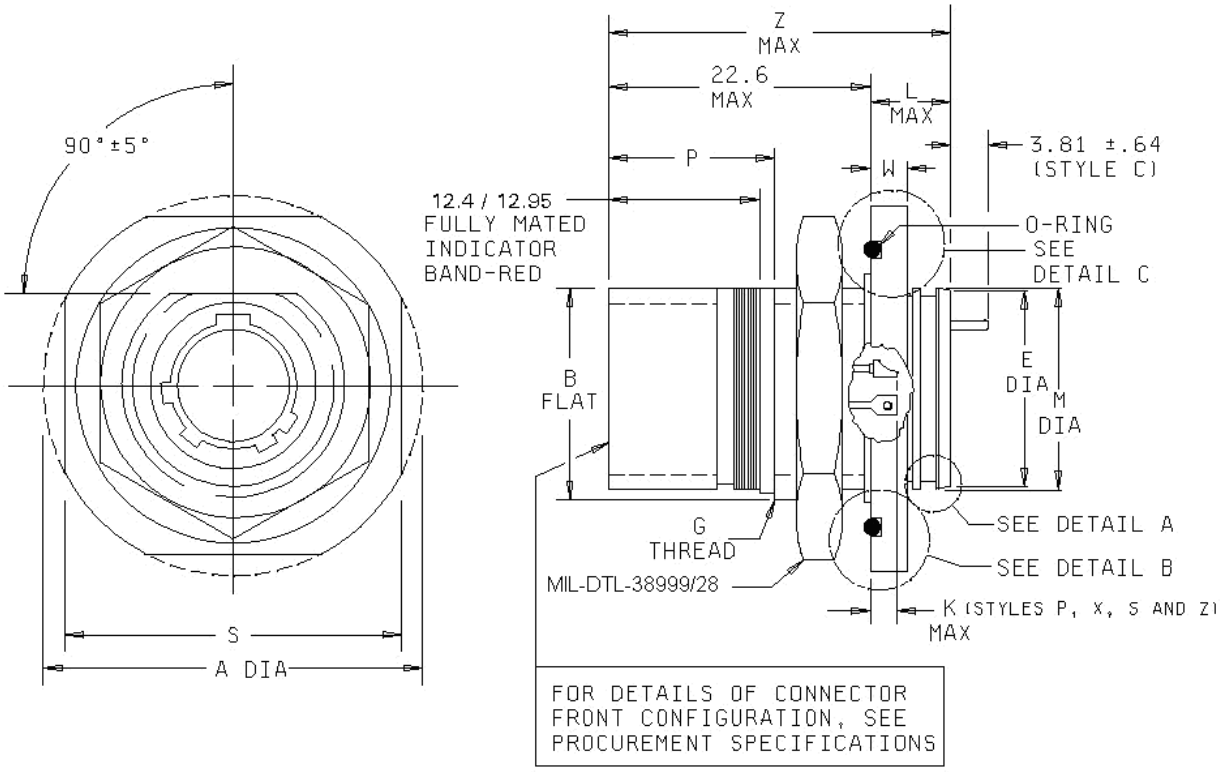
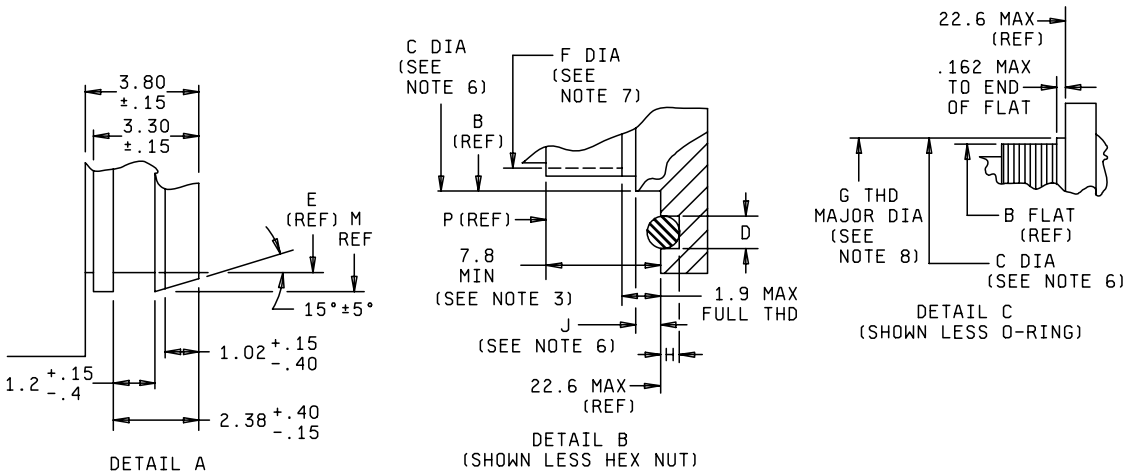


FIGURE 1. Receptacle, classes N and Y.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.



Connector dimensions

Shell size	Shell size code	A Dia ± .3	B Flat + .10 - .15	C Dia + .1 - .2	D + .13 - .03	E Dia ± .3	F Dia max	G thread
9	A	30.2 (1.189)	16.53 (.651)	17.4 (.685)	2.39 (.094)	15.3 (.602)	15.9 (.626)	M17X1-6g 0.100R
11	B	34.9 (1.374)	19.07 (.751)	20.6 (.811)	2.39 (.094)	18.4 (.724)	18.8 (.740)	M20X1-6g 0.100R
13	C	38.1 (1.500)	23.82 (.938)	25.4 (1.000)	2.39 (.094)	21.6 (.850)	23.8 (.937)	M25X1-6g 0.100R
15	D	41.3 (1.626)	26.97 (1.062)	28.5 (1.122)	2.39 (.094)	24.8 (.976)	26.8 (1.055)	M28X1-6g 0.100R
17	E	44.5 (1.752)	30.15 (1.187)	See note 6	2.39 (.094)	28.0 (1.102)	30.8 (1.213)	M32X1-6g 0.100R (see note 2)
19	F	49.2 (1.937)	33.32 (1.312)	See note 6	3.58 (.141)	31.2 (1.228)	33.8 (1.331)	M35X1-6g 0.100R
21	G	52.4 (2.063)	36.50 (1.437)	See note 6	3.58 (.141)	34.3 (1.350)	36.8 (1.449)	M38X1-6g 0.100R
23	H	55.6 (2.189)	39.67 (1.562)	41.2 (1.622)	3.58 (.141)	37.5 (1.476)	39.8 (1.567)	M41X1-6g 0.100R
25	J	58.7 (2.311)	42.85 (1.687)	44.4 (1.748)	3.58 (.141)	40.7 (1.602)	42.8 (1.685)	M44X1-6g 0.100R

FIGURE 1. Receptacle, classes N and Y- Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.

Shell size	Shell size code	H ± .13	J + .0 - .71	K max		L max
				With pin	With socket	
9	A	1.14 (.045)	1.5 (.059)	5.3 (.209)	5.9 (.232)	9.1 (.358)
11	B	1.14 (.045)	1.5 (.059)	5.3 (.209)	5.9 (.232)	9.1 (.358)
13	C	1.14 (.045)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.1 (.358)
15	D	1.14 (.045)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.1 (.358)
17	E	1.14 (.045)	---	5.1 (.201)	5.7 (.224)	9.1 (.358)
19	F	1.91 (.075)	---	5.1 (.201)	5.7 (.224)	9.7 (.382)
21	G	1.91 (.075)	---	5.1 (.201)	5.7 (.224)	9.7 (.382)
23	H	1.91 (.075)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.7 (.382)
25	J	1.91 (.075)	1.5 (.059)	5.1 (.201)	5.7 (.224)	9.7 (.382)

FIGURE 1. Receptacle, classes N and Y - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.

Shell size	Shell Size code	M Dia +.3, -.0	P +.4 -.1	S ±.4	W +.3 -.1	Z Max	Recommended O-ring SAE-AS3578 (see note 5)
9	A	16.3 (.642)	14.1 (.555)	27.0 (1.063)	2.6 (.102)	29.2 (1.150)	019
11	B	19.4 (.764)	14.1 (.555)	31.8 (1.252)	2.6 (.102)	29.2 (1.150)	022
13	C	22.7 (.894)	14.3 (.563)	34.9 (1.374)	2.6 (.102)	29.3 (1.154)	024
15	D	25.9 (1.020)	14.3 (.563)	38.1 (1.500)	2.6 (.102)	29.3 (1.154)	026
17	E	29.0 (1.142)	14.3 (.563)	41.3 (1.626)	2.6 (.102)	29.3 (1.154)	028
19	F	32.2 (1.268)	14.3 (.563)	46.0 (1.811)	3.4 (.134)	30.1 (1.185)	128
21	G	35.4 (1.394)	14.3 (.563)	49.2 (1.937)	3.4 (.134)	30.1 (1.185)	130
23	H	38.6 (1.520)	14.3 (.563)	52.4 (2.063)	3.4 (.134)	30.1 (1.185)	132
25	J	41.7 (1.642)	14.3 (.563)	55.6 (2.189)	3.4 (.134)	30.1 (1.185)	134

NOTES:
1. Dimensions are in millimeters.
2. Modified major diameter: 31.95 - 31.80 mm.
3. Panel thickness: 3.2 - 1.58 mm.
4. Polarizing position N is shown.
5. Alternate "0-ring" material may be SAE-AMS3325.
6. C and J dimensions define feature required for proper fit in type E panel.
7. F diameter defines thread start.
8. This dimension applies to sizes 17, 19, and 21 only (C diameter and G diameter are the same).
9. Inch equivalents are given for general information only.

FIGURE 1. Receptacle, classes N and Y - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1. Interface dimensions shall conform to MIL-DTL-38999.

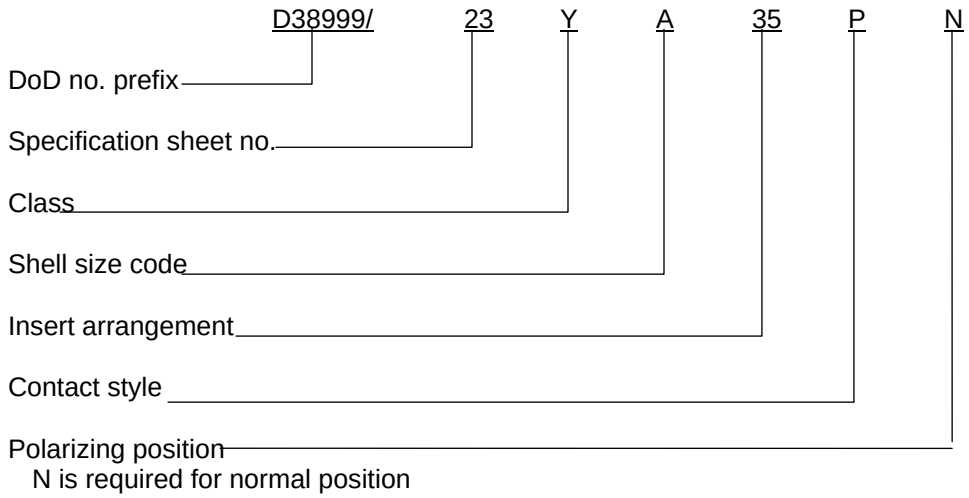
This connector mates with MIL-DTL-38999/26.

O-ring shall be provided with jam-nut receptacles.

For insert arrangements: See MIL-STD-1560.

Connectors accessories: See SAE-AS85049.

Part or Identifying Number (PIN) example:



N is required for normal position

Eyelet contact termination: See MIL-DTL-38999.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/28
MIL-DTL-38999/26
MIL-STD-1560
SAE-AS85049
SAE-AMS3325
SAE-AS3578

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2008-066)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T15:44Z
Check the source to verify that this is the current version before use.